

Preparation of porous activated carbon from semi-coke by high temperature activation with KOH for the high-efficiency adsorption of aqueous tetracycline

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ABSTRACT

Semi-coke activated carbon (SAC) were prepared from raw semi-coke (SC) by high temperature solid-state reaction with KOH and applied as an novel adsorbent to eliminate tetracycline (TC) from solution. The characteristics of SAC were dominantly affected by the mass ratio of KOH/SC and the pyrolysis temperature. As the mass ratio of KOH/SC increases, the specific surface area of SAC increased. As the pyrolysis temperature increases, the specific surface area of SAC increased firstly and then decreased. SAC prepared at optimum conditions at the pyrolysis temperature of 1073 K and mass ratio (KOH/SC) of 3:1 was found to be the specific surface area of $1365.79 \text{ m}^2\text{g}^{-1}$, the pore volume of $0.38 \text{ cm}^3\text{g}^{-1}$, and the average pore size of 3.1 nm. The adsorption capacity of TC increased with the increase of specific surface area and oxygen-containing functional groups, and the equilibrium adsorption capacity for TC on SAC-3 was 302.99 mg g^{-1} at 298 K. The high specific surface area, abundant surface functional groups, π - π bond and hydrophobic interaction between TC and SAC-3 contribute to good performance for TC adsorption. In addition, SAC-3 showed a good ability to resist cationic interference and good regeneration performance.

1. Introduction

China is a major country in the world that produces, uses and sales antibiotics [1,2]. With the acceleration of industrial development, the pollution of antibiotics has become more and more serious in recent years, which not only endangers human health, but also endangers the ecological environment. Tetracycline (TC) is widely used in production and life. After TC was used, every little TC is metabolized and most of the residual TC is discarded into the environment, which will accumulate in the environment [3]. The existence of TC in the aqueous environment poses a threat to human beings and our living environment [4].

Currently, the conventional methods for removing of TC from wastewater are as follows advanced oxidation processes [5], membrane filtration [6], electrochemical techniques [7], biotechnology [8], adsorption [9] and so on. Compared with other techniques, the adsorption technique has unique advantages because of its high efficiency, simplicity, safety and economic feasibility for eliminating TC from liquid waste [10–12]. The adsorption efficiency is determined by the type of adsorbent, pollutant characters, and the compositions of wastewater [13]. Most common adsorbents for removing TC such as carbon-based

materials, functionalized carbon nanotubes, graphene-based materials [14], bio-adsorbent [15] and so on. Among them, activated carbon (AC) has been extensively used because of their high specific surface areas, abundant porous framework, and adjustable surface functionality [16,17]. Nonetheless, AC with the high cost and poor regeneration performance limits the application of AC in the wastewater treatment field. Therefore, the development of cost-effective AC is urgent.

Recently, semi-coke (SC) The superior properties of semi-coke (SC) include variety of functional groups and high surface area has received widespread attention of domestic and foreign Scholars [18]. SC is one kind of solid waste, which is cheap, easy to get and well-developed pore structure [19]. The SC have a developed microporous structure and an irregularly arranged microcrystalline layer sheet structure, which is rich in C, H, and O elements, and most of the oxygen atoms, hydrogen atoms and carbon atoms are combined by the chemical bond of hydroxyl, ethoxy and carbonyl groups. However, raw SC exhibits a relatively low adsorption capacity for organic or inorganic contaminants because of the existence of tar and ash on its surface [20]. Hence, the exploration of physicochemical activation methods to improve the surface structure of SC has received widely attention.

The chemical activation method is to mix the SC powder with the

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Nomenclature

Acronyms and abbreviations

SC	semi-coke
AC	activated carbon
SAC	semi-coke activated carbon
V	volume of solution (mL)
M	mass of adsorbent (mg)
K_f	Freundlich constant
k_1	rate constant of the first-order kinetic model

	($\text{g}\cdot\text{mg}^{-1}\cdot\text{min}^{-1}$).
n	Freundlich constant
t	time (min)
TC	tetracycline
C_0	tetracycline initial concentration ($\text{mg}\cdot\text{L}^{-1}$)
C_e	tetracycline equilibrium concentration ($\text{mg}\cdot\text{L}^{-1}$)
S_{BET}	volume of specific surface area
q_e	adsorption capacity of TC in equilibrium ($\text{mg}\cdot\text{g}^{-1}$)
q_m	maximum adsorption capacity ($\text{mg}\cdot\text{g}^{-1}$)
k_2	rate constant of the second-order kinetic model (1/min).
b	affinity coefficient ($\text{L}\cdot\text{mg}^{-1}$).

activated agent and then activate it at a suitable temperature to improve the specific surface area of SC. There are activating agents most used such as HNO_3 [21], ZnCl_2 [22], KOH [23,24], K_2CO_3 [25], etc. After chemical activation, SAC has excellent pore structure and surface-active groups, which not only improved good adsorption for waste gas and waste water, but also has a large range of raw material resource, low cost, simple production process and good regeneration performance. For example, SAC prepared via KOH activation exhibits larger surface area ($347.39 \text{ m}^2\cdot\text{g}^{-1}$) than AC by the conventional treatment ($3.19 \text{ m}^2\cdot\text{g}^{-1}$), which shows the maximum adsorption capacity of $42.75 \text{ mg}\cdot\text{g}^{-1}$ for aqueous phenol [20]. Zhu et al. [21] reported that the porosity of biochar can be further improved and the adsorption capacity of biochar to antibiotics in water can be improved in the mixed gas-bearing environment of wood at 873 K–1073 K.

In this work, SAC was prepared by high temperature solid state activation with KOH , and the activation conditions including the mass ratio of KOH/SC and the pyrolysis temperature was studied. The surface properties, constituent elements, and text features of SAC were systematically characterized by BET, XPS, FT-IR, SEM and Raman. TC adsorption behaviors on SAC were investigated in detail, including adsorption capacity under different activation conditions and initial TC concentrations, as well as adsorption kinetics, equilibrium, cation effects, and the adsorption mechanism was also advised.

2. Materials and methods

2.1. Reagents

Tetracycline hydrochloride (Biotechnology level) was brought from Shanghai Macklin biochemical technology Co., Ltd. KOH (AR, 99%) and other reagents were acquired from Tianjin Kemiu Chemical Reagent Co., Ltd. The SC was from Fugu Wanxing New Energy Co. Ltd (Yulin, China).

2.2. Preparation of semi-coke activated carbons (SAC)

The raw SC is washed three times with de-ionized water, and the fine powder in the SC of the raw material is removed, and then dried in oven at 378 K until constant weight. By grinding the dry SC into particles smaller than 20 mesh, and grind SAC to KOH in a certain proportion (SC/KOH mass ratio 1:0–3). Then the mixture was treated in a tubular resistance furnace at 773 K–1173 K under nitrogen protection ($80 \text{ mL}/\text{min}$) at a heating rate of $283 \text{ K}\cdot\text{min}^{-1}$. After calcinated for 3 h, the obtained SAC was repeatedly washed with deionized water until to neutral, and dried at 378 K for 24 h. (denoted as SAC-x; x delegates mass ratio of KOH/SC , and the SAC after TC adsorption is denoted as ASAC).

2.3. Characterization

Surface analysis of synthesized samples was characterized by X-ray photoelectron spectrum (XPS) (AXIS SUPRA, UK). The surface

functional groups were described by FT-IR spectra from 400 to 4000 cm^{-1} with KBr techniques (VECTOR-22, Bruker, Germany). The N_2 adsorption-desorption isotherms of the samples were collected at 77 K on a Micromeritics ASAP 2460 apparatus, and the BET specific surface area was determined. Raman spectra were conducted with a Laser Resonant Raman-transient fluorescence spectrometer (Lab RAM HR Evolution, S.A.S.) using a 532 nm laser as an excitation source. The surface morphology of the samples was described by scanning electron microscope (SEM) (Quanta 400 FEG, US).

2.4. Adsorption experiments

A series of adsorption experiments were conducted in brown glass tube receiving of 50 mL of reaction solution. Thirty milligrams of SAC-3 were put into different concentrations aqueous solution of TC at pH 6.0. Then the mixtures were shake at 140 rpm and 298 K in bath oscillator for 6 h. For kinetic study, 300 mg of SAC-3 was dispensed in 500 mL three-necked flask receiving 500 mL of TC solution with the initial concentration of 90, 120, 150 $\text{mg}\cdot\text{L}^{-1}$, and adjust the pH of the system to 6.0 at 298 K for 8 h. After adsorption equilibrium, the filtrate was filtered using a $0.45 \mu\text{m}$ acetate filter, and the absorbance was detected by an ultraviolet–visible (UV–Vis) spectrophotometer at a wavelength of 356 nm. The adsorption amount, q_e (mg/g), was determined as follows:

$$q_e = \frac{(C_0 - C_e)V}{M} \quad (1)$$

where C_0 ($\text{mg}\cdot\text{L}^{-1}$) and C_e ($\text{mg}\cdot\text{L}^{-1}$) are the initial and equilibrium concentration of TC in solution, respectively. M (mg) is the mass of adsorbent and V (L) is the volume of solution.

For the regeneration experiments, TC saturated adsorbents were treated with NaOH solution ($0.2 \text{ mol}\cdot\text{L}^{-1}$) for 12 h at 298 K. The regenerated adsorbents were separated and washed with deionized water until neutral for reuse. The regenerated experiments were repeatedly for 3 times.

In addition, for TC adsorption in four real water system (ultrapure water, tap water, lake water, and discharged sewage) on the adsorbent, 500 mg of SAC-3 was added and stirred in 200 mL solution of initial TC concentrations of 50 and $150 \text{ mg}\cdot\text{L}^{-1}$ at pH 6.0 and 298 K.

2.5. Experimental data models

The data for TC adsorption by SAC-3 was analyzed by the Langmuir and Freundlich isotherm model.

The Freundlich model is used to analyze the adsorbents (2):

$$q_e = K_f C_e^{1/n_f} \quad (2)$$

The Langmuir model are as follows (3):

$$q_e = \frac{q_m b C_e}{1 + b C_e} \quad (3)$$

where q_e ($\text{mg}\cdot\text{g}^{-1}$) is the adsorption amount of TC at equilibrium, C_e

($\text{mg}\cdot\text{L}^{-1}$) is the equilibrium concentration of TC, K_f and n_f are the characteristic constant of Freundlich, q_m ($\text{mg}\cdot\text{g}^{-1}$) is the maximum adsorption capacity of TC and b ($\text{L}\cdot\text{mg}^{-1}$) is affinity coefficient.

The adsorption kinetic model (pseudo-first-order kinetic model and pseudo-second-order kinetic model) was used to simulate experimental data to study the adsorption rate of TC by SAC. The first-order kinetic model can be used to explain the adsorption rate of organic pollutants on the adsorbents, which can be expressed as:

$$q_t = q_e (1 - e^{-k_1 t}) \quad (4)$$

The calculation formula of pseudo-second-order kinetic equation is:

$$q_t = \frac{k_2 q_e^2 t}{1 + k_2 q_e t} \quad (5)$$

where q_e and q_t ($\text{mg}\cdot\text{g}^{-1}$) was the equilibrium adsorption capacity, and the adsorption amount at time t (min), k_1 (min^{-1}) and k_2 ($\text{g}\cdot\text{mg}^{-1}\cdot\text{min}^{-1}$) represent the rate constants of two kinetic models, respectively.

3. Results and discussion

3.1. Influence of pyrolysis condition on the preparation of SAC

Mass ratio of KOH/SC and pyrolysis temperature is the main factors which influence the pore structure properties of SAC and adsorption efficiency. The influence of pyrolysis temperature (773 K–1173 K) and mass ratio of KOH/SC (0–3) on the specific surface area of SAC were illustrated in Fig. 1(a). From the results, the specific surface area of SAC augmented as the mass ratio of KOH/SC increased, and the optimum pyrolysis was found to be at 1073 K [26]. The maximum specific surface area is $1365.79 \text{ m}^2\cdot\text{g}^{-1}$ at pyrolysis temperature of 1073 K and mass ratio of KOH/SC of 3:1.

The activation of SC by KOH is realized by the chemical activation between alkali and hot carbon [27]. In the activation process, KOH penetrates further into the pore structure of the SC and is eventually converted into other materials and retained in the pores. After the reaction finished, the residual chemical reagents and impurities in the SAC were washed away to obtain abundant pores. With the increasing pyrolysis temperature, the iodine value and the surface area of SAC increases first and then decreases, reaching a peak at 1073 K (Fig. 1(b)). The high temperature was favorable for the rapid increase of the surface area during the KOH activation process. This is because when the pyrolysis temperature is below 1073 K, as the pyrolysis temperature increases, SC volatiles and oxygen groups decompose, resulting in an increase of basic functional groups, forming more microporous structures, which improved the activity and adsorption properties of SAC. However, when the pyrolysis temperature is above 1073 K, the surface area began to decrease, which may be because that the consumption of KOH increases with the increasing pyrolysis temperature, leading to the relative decrease of the final effect of KOH and the collapse of the internal structure of SAC.

With the increase of mass ratio of KOH/SC, the iodine value and the specific surface area of SAC were increased (Fig. 1(c)), which could be due to the KOH enters the original pores, or affected by the gasification of the pore walls, causing large pores to form between adjacent pores. In conclusion, the iodine value changes as well as the specific surface area improve with the increasing mass ratio of KOH/SC, and increase first and then decrease with the increase of pyrolysis temperature, which may be caused by the augmented active sites of large specific surface area.

3.2. Characterization of SC and SAC

The surface structure of SC and SAC was detected by N_2 adsorption measurements at 77 K. The N_2 adsorption-desorption isotherms were

shown in Fig. 2(a). The N_2 adsorption amount of SAC is much higher than SC. Based on IUPAC adsorption isotherm classification, the isotherm of SC belongs to type III, which indicates that there is a weak interaction force between SC and N_2 , and the existence of various pores on the surface of SC is negligible. As seen Fig. 2(a), that all of SAC curves were type IV isotherms typically with a type H4 hysteresis, due to the presence of mesoporous and microporous. [17]. SAC-3 is discriminated among the synthesized materials by the largest specific surface area ($1365.79 \text{ m}^2\cdot\text{g}^{-1}$).

The pore size distribution curves of SC and SAC are given in Fig. 2(b). The pore texture characteristic was very similar for SAC samples in terms of the pore volume, and pore size distribution. As tabulated in

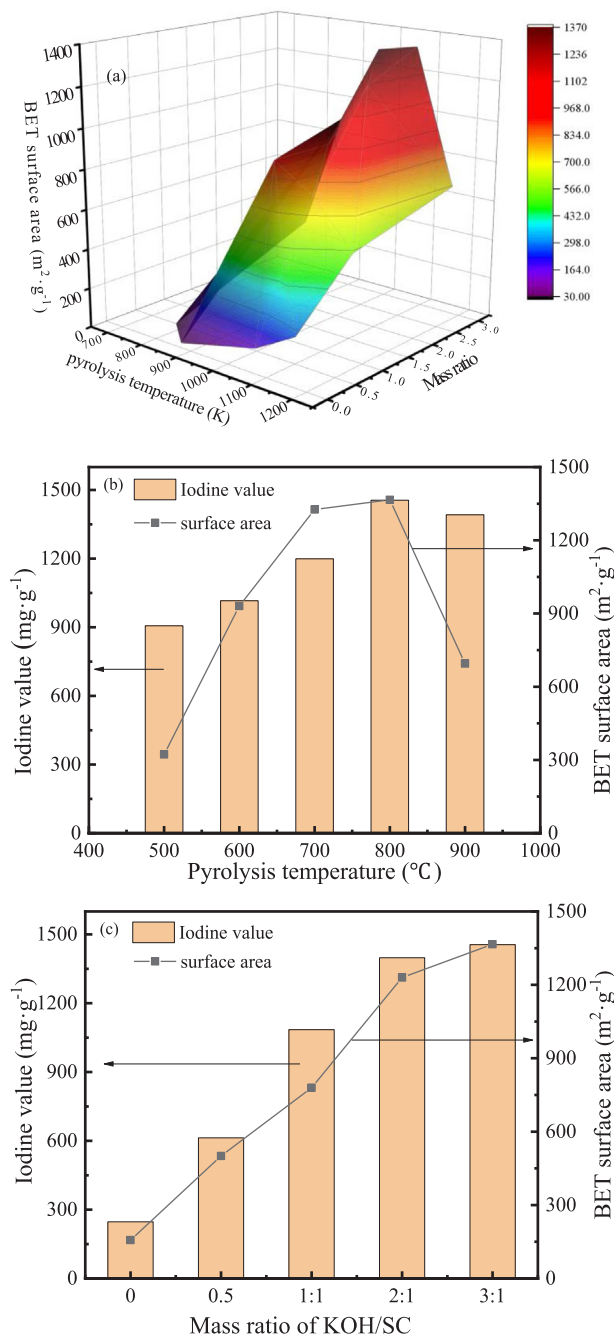


Fig. 1. Effect of mass ratio of KOH/SC and pyrolysis temperature on the specific surface area of adsorbents (a), and the influence of pyrolysis temperature (b) and mass ratio of KOH/SC (c) on the iodine value and the specific surface area of the adsorbent.

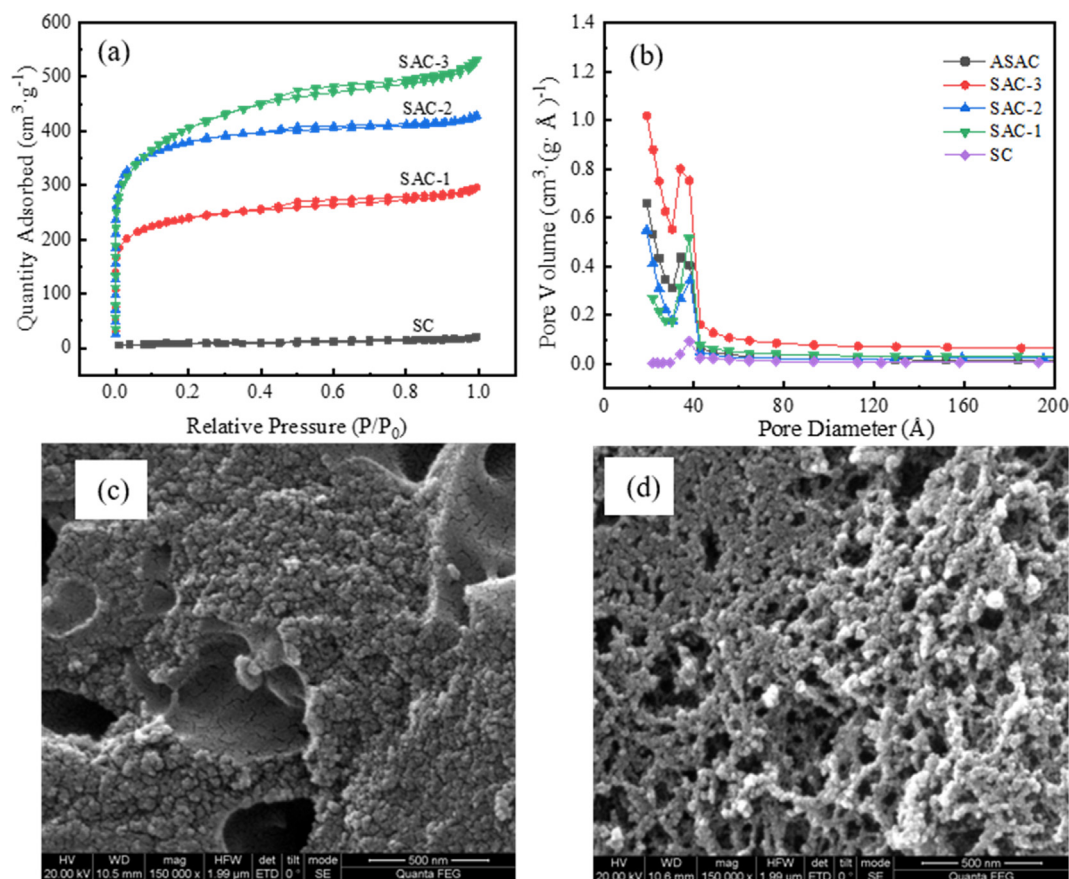


Fig. 2. N_2 adsorption-desorption isotherms (a) and the pore size distribution (b) of SAC, SC before and after TC adsorption, the SEM images of SC (c) and SAC-3 (d).

Table 1

Porosity characteristics of the SAC and SC.

Sample	q_m^a ($mg \cdot g^{-1}$)	S_{BET} ($m^2 \cdot g^{-1}$)	BJH desorption average pore diameter (nm)	BJH pore volume b^b ($cm^3 \cdot g^{-1}$)
SC	30.71	27.22	5.5	0.02
SAC-1	167.29	779.69	4.0	0.13
SAC-2	232.87	1229.61	3.0	0.15
SAC-3	315.12	1365.79	3.1	0.38
ASAC	–	1073.92	2.8	0.20

^a Fitting results of tetracycline adsorption data with the Langmuir model (Experimental conditions: $T = 298$ K, $pH = 6.0$).

^b Total pore volume determined at $P/P_0 = 0.99$

Table 1, in most cases, the total pore volume and the mean pore width depend on the amount of KOH used, and all produced activated carbons are essentially mesoporous [28]. The average pore size of SAC-3 is 3.1 nm, and pore volume about $0.38 \text{ cm}^3 \cdot g^{-1}$. KOH activation can make micropores increased, the average pore size decreased, which indicating that the activation of SC by KOH activated method formed a porous structure and improved structural properties of SC.

The morphologies of SC and SAC-3 were studied with SEM as shown in Fig. 2 (c) and (d). In Fig. 2 (c), the surface of the SC was smooth except for some pores, and the diameter of pores in SC is higher than SAC-3, the porous structure was further certified by pore size distribution. As illustrated in Fig. 2 (d), there are many uniform small pores on the surface of SAC-3, which may be because that the KOH could eliminate the ash and tar on the surface of the SC and release of small molecules during activation, such as CO_2 , H_2 , CH_4 . These pores may provide an enter close for the migration of TC molecules onto the active sites on SAC-3 and the large specific surface area, thus contributing to

the enhanced TC adsorption [29].

The FT-IR spectra of the SAC and SC samples are shown in Fig. 3 (a). The four adsorbents exhibit many absorption peaks at the range of 400 to 4000 cm^{-1} . The results of FT-IR spectra SAC-1, SAC-2 and SAC-3 were similar. The absorption peak at 3460 cm^{-1} is the stretching vibration of the O–H bond [30]. The absorption peak at 1610 cm^{-1} is ascribed to the C=C stretching in the aromatic structure, the C=O stretching of the COO^- or COO^- conjugated aromatic C=C [30]. The absorption peak at 1353 cm^{-1} is the stretching vibration of phenolic [31]. The absorption peak at 1141 cm^{-1} is attributed to the C–O bond tensile vibration [32], and the absorption peak at 1000 cm^{-1} may be ascribed to C–H tensile vibration of substituted aromatic or ash [33], which suggested that SC was successfully activated by KOH.

The Raman spectra of SC and SAC are shown in Fig. 3 (c). There are two typical characteristic peaks of SAC: D peak and G peak, where in the D peak appearing at 1332 cm^{-1} ascribed to the in-plane vibration of sp^2 -bonded carbon structure, and the G band at 1582 cm^{-1} corresponded to the in-plane vibration of sp^2 -bonded graphitic carbon structure [34]. The obvious change of two characteristic peaks of SC and SAC was observed, indicating of the change of carbon structures during activation [35].

The elemental compositions of the SC and SAC were evaluated by XPS analysis. The high-resolution C1s XPS spectra are presented in Fig. 4. The C1s peak binding energy centered on 284.6 eV is used as a calibration standard, and C peak is convoluted into four characteristic peaks: aliphatic/aromatic group (C–C/C–H, 284.6 eV), hydroxyl group (C–O, 285.8 eV), carbonyl group (C=O, 287.0 eV), ester or lactone group (O=C–O, 289.5 eV) in alcohols or phenols [36]. The binding energy of the peaks and atomic ratio are tabulated in Table 2. From the results, KOH activation decreased the atomic ratio of C–C/C–H groups and increased the total atomic ratio of C=O and O=C–O groups. Moreover, with the increased mass ratio of KOH/SC, the atomic ratio of

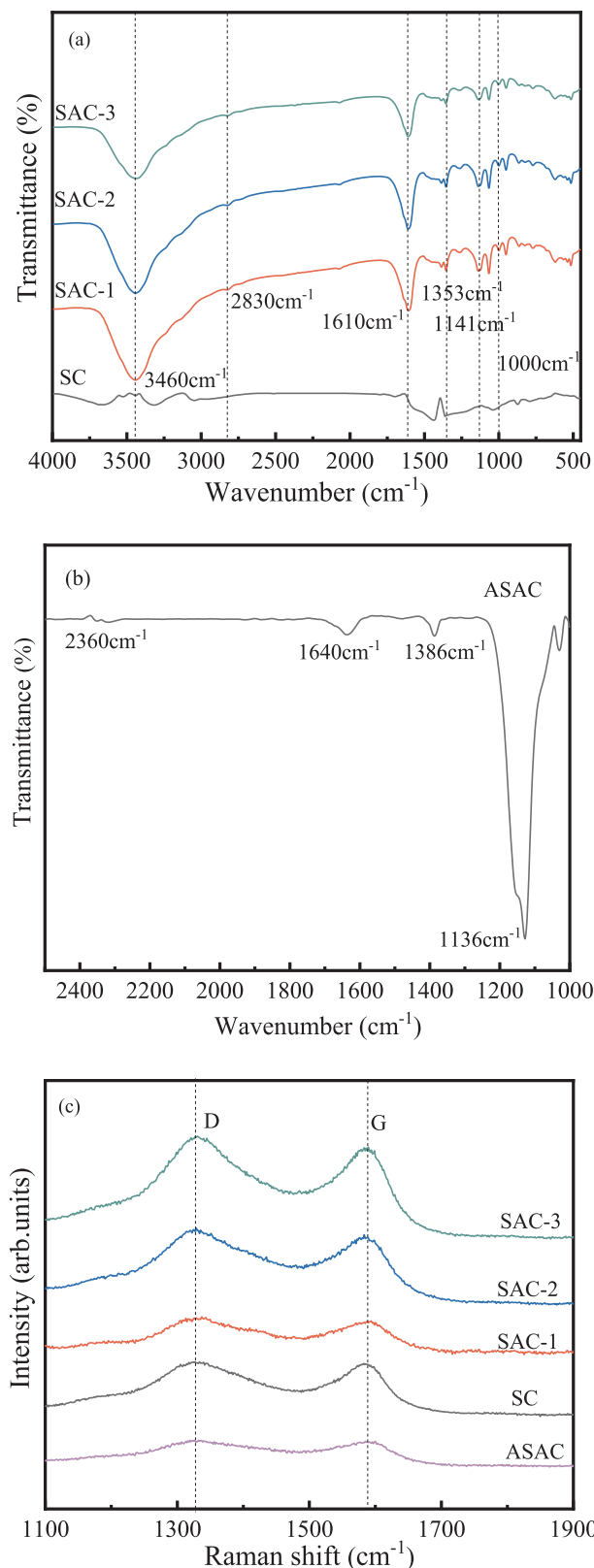


Fig. 3. FT-IR spectra of SC and SAC (a) and SAC-3 after TC adsorption (b), Raman spectra of SC, SAC before and after TC adsorption (c).

C–C/C–H groups decreased and the total atomic ratio of C=O and O=C–O groups augmented. FT-IR and XPS confirmed that oxygen-containing functional groups such as COO[−], C=O and C–O existed on the surface of the SAC.

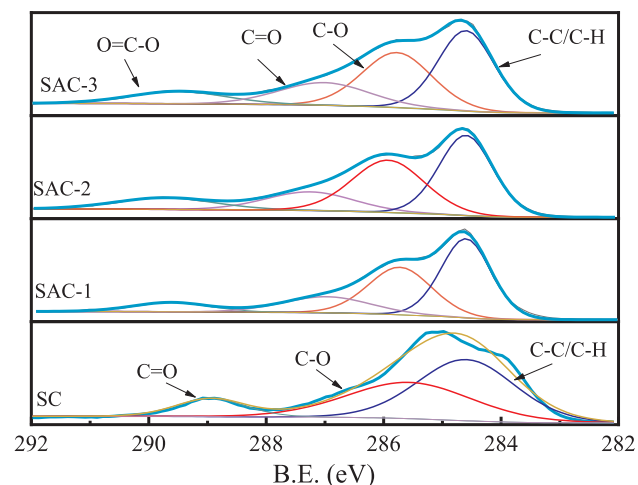


Fig. 4. XPS spectra of C1s peak for SC and SAC.

Table 2

Deconvolution results of C1s peaks for SC and SAC.

B.E. (eV)				Relative contents (%)				Assignment
SC	1:1	2:1	3:1	SC	1:1	2:1	3:1	
284.60	284.60	284.60	284.60	50.76	43.47	39.74	38.82	C–C/C–H
285.56	285.72	285.92	285.78	38.68	33.13	34.68	33.14	C–O
288.94	286.97	287.25	287.02	10.57	15.55	15.04	17.77	C=O
–	289.62	289.72	289.50	–	7.85	10.54	10.27	O=C–O

3.3. Adsorption of TC

3.3.1. Adsorption isotherms

The adsorption isotherm was used to describe the interaction between the adsorbent and the adsorbate. As shows in Fig. 5, the isotherm adsorption curves of TC on SC and SAC-3 at 298 K and pH 6.0, the maximum equilibrium adsorption capacity was 302.99 mg·g^{−1}. From the results, the TC adsorption amount increased remarkably after KOH activation, and TC adsorption increased with the increasing mass ratio of KOH/SC, which is similar with the tendency of the specific area and the C=O and O=C–O groups, representing that the specific area and the oxygen containing groups play the main role for TC adsorption on SAC-3. The pH value of the zero-charge point (pH_{zpc}) was measured by the pH drift method [37]. The pH_{zpc} of SAC-3 was found to be around 8.1, indicating that the surface of SAC-3 was positively charged over the pH range. The high surface area greatly promotes the adsorption of TC onto SAC-3 and gives it abundant active sites for electrostatic interaction and hydrogen bonding.

The fitting results are described in Table 3. It was found that the fitting of Langmuir isotherm equation (0.956) fit adsorption process of TC better than that of Freundlich isotherm equation (0.921), which suggested that TC adsorption process was monolayer adsorption. The potential driving force for TC to adsorb on SAC-3 might be the H-bonding between surface active groups of SAC-3 and the TC molecules [38]. The adsorption mechanism is probably ascribed to the π – π bond interaction and hydrophobic interaction between TC and SAC-3 [4,39].

3.3.2. Adsorption kinetics

Adsorption kinetics of TC on SAC-3 at the initial concentrations of 90, 120, 150 mg·L^{−1} at 25 °C were illustrated in Fig. 6. From the results, the fast TC adsorption on SAC-3 was found in the first 50 min, then the adsorption rate gradually became slow and TC adsorption on the adsorbent reached equilibrium after 6 h. The slow adsorption rate after 50 min may be due to that TC molecules enter and penetrate the mesopores of SAC-3, which can also be confirmed by the reduced average

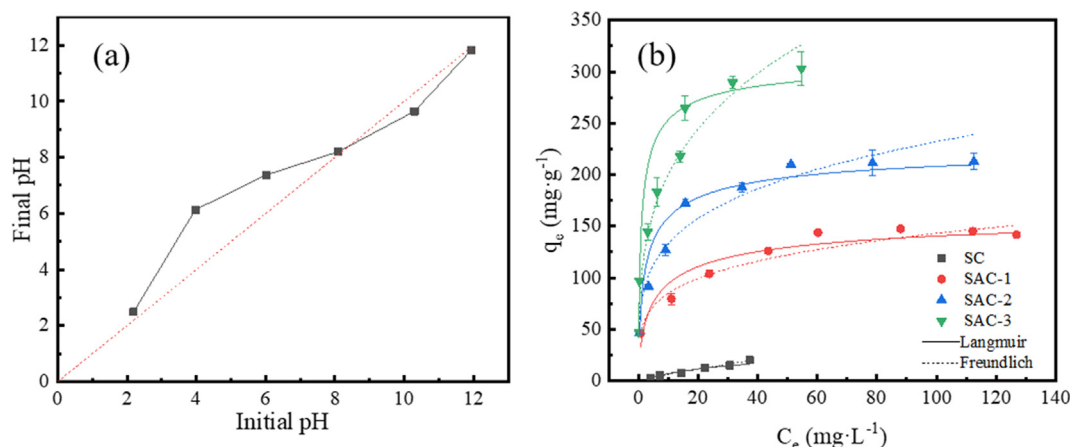


Fig. 5. pH drift method to obtain the pH_{ZPC} (a). and adsorption isotherms of TC adsorption on SC and SAC and the corresponding nonlinear fitting by Langmuir and Freundlich models(b). (Experimental conditions: $C_0 = 30\text{--}240 \text{ mg}\cdot\text{L}^{-1}$, $m = 30 \text{ mg}$, $V = 50 \text{ mL}$, $T = 298 \text{ K}$, $\text{pH} = 6.0$ and $t = 6 \text{ h}$).

Table 3

Isotherm parameters for the TC adsorption on adsorbents. (Experimental conditions: $C_0 = 30\text{--}240 \text{ mg}\cdot\text{L}^{-1}$, $m = 30 \text{ mg}$, $V = 50 \text{ mL}$, $T = 298 \text{ K}$, $\text{pH} = 6.0$ and $t = 6 \text{ h}$).

Samples	Langmuir			Freundlich		
	$q_m (\text{mg}\cdot\text{g}^{-1})$	$b (\text{L}\cdot\text{mg}^{-1})$	R^2	n	$K_f (\text{mg}\cdot\text{g}^{-1})$	R^2
SC	30.71	0.030	0.925	1.17	0.88	0.990
SAC-1	167.29	0.320	0.954	4.44	50.63	0.898
SAC-2	232.87	0.548	0.978	4.17	76.80	0.975
SAC-3	315.12	0.974	0.956	3.49	103.69	0.921

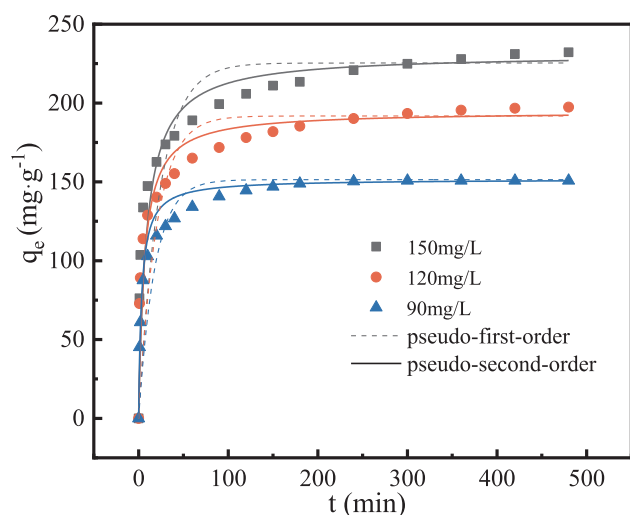


Fig. 6. Adsorption kinetics of TC adsorption on SAC-3. (Experimental conditions: $C_0 = 90\text{--}150 \text{ mg}\cdot\text{L}^{-1}$, $m = 300 \text{ mg}$, $V = 500 \text{ mL}$, $T = 298 \text{ K}$, and $t = 5\text{--}480 \text{ min}$).

Table 4

Adsorption kinetics parameters of TC on SAC-3. (Experimental conditions: $C_0 = 90\text{--}120 \text{ mg}\cdot\text{L}^{-1}$, $m = 300 \text{ mg}$, $V = 500 \text{ mL}$, $T = 298 \text{ K}$, $\text{pH} = 6.0$ and $t = 5\text{--}480 \text{ min}$).

$C_0 (\text{mg}\cdot\text{L}^{-1})$	$q_{eq} (\text{mg}\cdot\text{g}^{-1})$	Pseudo-first-order kinetics			Pseudo-second-order kinetics		
		$k_1 (\text{min}^{-1})$	$q_{cal} (\text{mg}\cdot\text{g}^{-1})$	R^2	$k_2 (\text{min}^{-1})$	$q_{cal} (\text{mg}\cdot\text{g}^{-1})$	R^2
90	150.77	0.055	151.42	0.749	0.0017	149.08	0.977
120	197.39	0.050	191.79	0.612	0.0015	186.64	0.932
150	232.21	0.043	225.38	0.589	0.0007	226.53	0.896

pore diameter and pore volume of ASAC after TC adsorption in Table 1.

The corresponding results were drawn in Table 4. The correlation coefficient (R^2) of the pseudo-second-order kinetic model at initial TC concentrations of 90, 120, 150 $\text{mg}\cdot\text{L}^{-1}$ is higher than that of the first-order kinetic model, and the theoretical maximum adsorption amount fitted by the pseudo-second-order kinetic is similar to the experimental test results, indicating that the adsorption kinetics of TC by SAC-3 follows the pseudo-second-order reaction equation. The adsorption rate constants of TC adsorption on the adsorbent decreased with the initial TC concentration increases, which may be because the concentration gradient of TC provides the driving force for the TC adsorption [40,41].

3.3.3. Effect of solution pH and cations on TC adsorption

The changes of solution pH not only influence the variation of surface function group, but also affect the existence of TC ions in the waste water. As shown in Fig. 7(a), TC adsorption on SAC-3 at pH in the range of 3–11 and 298 K. The pH of the solution has a great influence on the removal of TC on SAC-3. As solution pH is less than 5.0, TC adsorption on SAC-3 increases as solution pH augments. The adsorption capacity of TC on SAC-3 decrease with the increase of pH value above 5.0, while the maximum adsorption capacity of SAC-3 for TC was found to be at pH 5.0. TC molecules with multiple functional groups form many species at different pH that exist predominantly as amphoteric ions in the typical pH in the natural aqueous environment ($\text{pK}_{a1} = 3.42$, $\text{pK}_{a2} = 7.59$, $\text{pK}_{a3} = 9.00$) [9,42]. For example, TC mainly exists as cation (+, 0, 0) at pH below 3.4, as zwitterion (+, −, 0) with pH between 3.4 and 7.6, as an anionic species (+, −, −) with pH above 7.6. When the solution is alkaline, TC in the solution mainly exists in the form of an anion, and the oxidized group on the surface of the SAC-3 is gradually ionized, and the surface charge density and the amount of adsorption decrease as the pH increases, which may be because an electrostatic repulsion between the TC and active chemical groups of the SAC-3 decreased TC adsorption on the SAC-3. Therefore, the electrostatic attraction between TC and the SAC-3 plays a key role

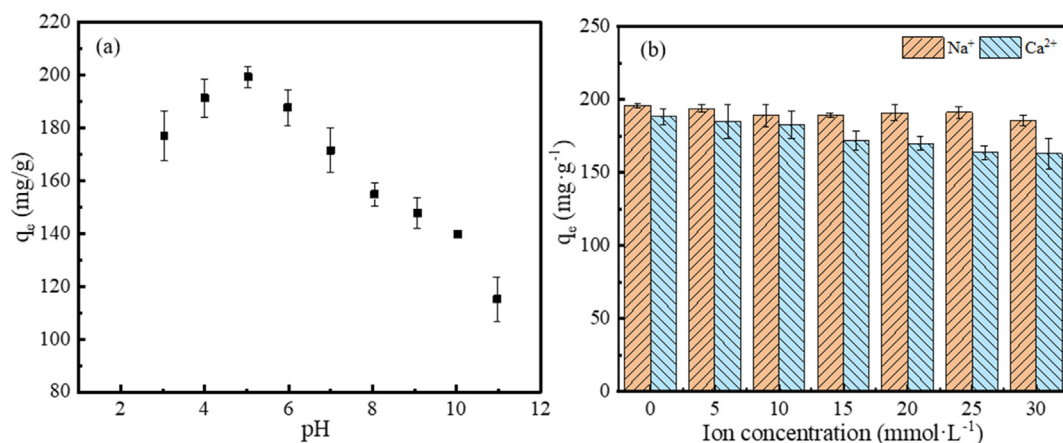


Fig. 7. Effect of solution pH (a) (Experimental conditions: $C_0 = 120 \text{ mg}\cdot\text{L}^{-1}$, $m = 30 \text{ mg}$, $V = 50 \text{ mL}$, $T = 298 \text{ K}$, $\text{pH} = 3\text{--}11$ and $t = 6 \text{ h}$) and cations (b) (Experimental conditions: $C_0 = 120 \text{ mg}\cdot\text{L}^{-1}$, $m = 30 \text{ mg}$, $V = 50 \text{ mL}$, $T = 298 \text{ K}$, $\text{pH} = 6$, $\text{Na}^+ = 0\text{--}30 \text{ mmol}\cdot\text{L}^{-1}$, $\text{Ca}^{2+} = 0\text{--}30 \text{ mmol}\cdot\text{L}^{-1}$ and $t = 6 \text{ h}$).

during the adsorption process [43].

The effects of inorganic cations (Na^+ , Ca^{2+}) may interfere in the adsorption of TC onto the adsorbent (Fig. 7 (b)) [44]. From the calculating results, the presence of Na^+ exhibits slight effect on the TC removal, and TC adsorption on SAC-3 decreased with the increasing concentration of Ca^{2+} . The decreased TC adsorption on the adsorbent in presence of cations may be caused by the competition between the cations and TC for the active sites of the SAC-3.

3.3.4. Regeneration of adsorbents

The regeneration of the adsorbent is an important index to evaluate the efficiency of adsorbent [45]. The adsorption performance of TC on regenerated SAC-3 was presented in Fig. 8. The capacity of TC adsorbed on the regenerated adsorbent reduced, which may be ascribed to the incomplete desorption of TC from SAC-3 [35]. After three cycles of desorption-regeneration, the adsorption capacity of the regenerated SAC-3 for TC was maintained at $115.10 \text{ mg}\cdot\text{g}^{-1}$, indicating that the SAC-3 had good reuse performance and can be used a potential candidate material for TC adsorption in water.

3.3.5. Real wastewater application

The complex composition of actual wastewater will affect the removal rate of TC by adsorbent, so it is need to investigate the application of SAC-3 in actual wastewater [15]. The removal of TC by SAC-3 in ultrapure water, tap water, lake water, and discharged sewage was shown in Fig. 9. From the results, when the concentration of TC was $50 \text{ mg}\cdot\text{L}^{-1}$, TC in different water was almost completely removed by SAC-3, which may be because the adsorbent can provide the enough active sites for TC and coexisting organic and inorganic matter. However, at the TC concentration of $150 \text{ mg}\cdot\text{L}^{-1}$, the removal rate of TC from municipal sewage and lake water was reduced, which may be possibly because that a large amount of organic contaminants coexisting in the wastewater will occupied the active sites of SAC-3 by competition with TC for the active sites of the adsorbent, which resulted in a reduced TC adsorption in municipal wastewater. These results show that SAC-3 adsorbents have great potential for treating actual wastewater, and can work effectively under complex conditions.

3.4. Adsorption mechanism

Compared with SAC-3, SAC-3 after TC adsorption exhibits the reduced specific surface area, the average pore size and pore volume, reflecting that TC was adsorbed and penetrated on the mesopore of the SAC-3. From the results of the FT-IR spectra in Fig. 3(b), the peak at 3460 cm^{-1} of the SAC-3 after TC adsorption disappeared, suggesting

that hydroxylic groups of the adsorbent are involved in the π - π bond interactions with TC molecules [46]. The obvious peaks at 1640 , 1386 and 1136 cm^{-1} was observed for SAC-3 after TC adsorption, and the peak at 1141 cm^{-1} shifted to 1136 cm^{-1} , reflecting that the stretching vibration of $-\text{OH}$ or $\text{C}-\text{O}$ groups in carboxylate or ethers contributed to the TC adsorption on SAC-3 (Fig. 3 (b)). Moreover, the peak of $\text{C}=\text{O}$ shifted from 1640 cm^{-1} to 1610 cm^{-1} , suggesting that π - π bond interactions may contribute to the enhanced TC adsorption on the adsorbent [47]. Moreover, the peak of SAC-3 after TC adsorption shifted from 2830 cm^{-1} to 2927 cm^{-1} , indicating of the developing of hydrophobic interaction between the amide group in TC molecules and the surface-active groups of SAC-3 [48]. Fig. 3 (c) shows the Raman spectra of SAC-3 after TC adsorption. The intensity of the D peak and G peak of SAC-3 after TC adsorption sharply decreased, indicating that the carbon structure transformation occurred in TC adsorption process.

4. Conclusions

SAC was successfully prepared via high temperature activation with KOH. The results indicate that the BET specific surface area of SAC increased with the increase of mass ratio of KOH/SC. As the pyrolysis temperature increases, the specific surface area of SAC first increases and then decreases. The optimum activation conditions were determined as the temperature was 1073 K and the mass ratio of KOH/SC

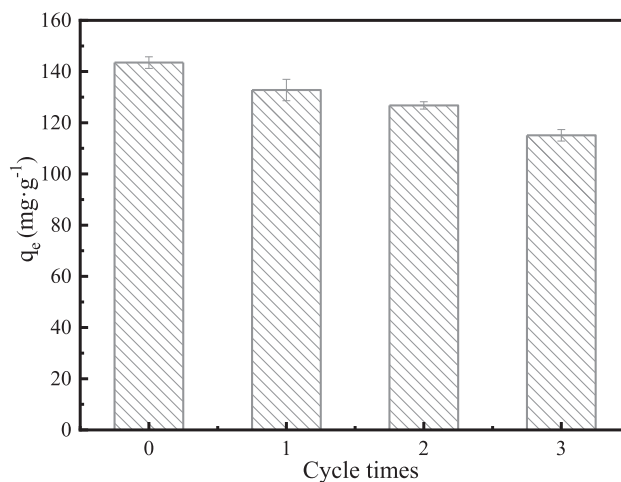


Fig. 8. Adsorption amounts of TC on the regenerated SAC-3. (Experimental conditions: $C_0 = 100 \text{ mg}\cdot\text{L}^{-1}$, $m = 30 \text{ mg}$, $V = 50 \text{ mL}$, $T = 298 \text{ K}$, $C_{\text{NaOH}} = 0.2 \text{ mol}\cdot\text{L}^{-1}$ and $t = 12 \text{ h}$).

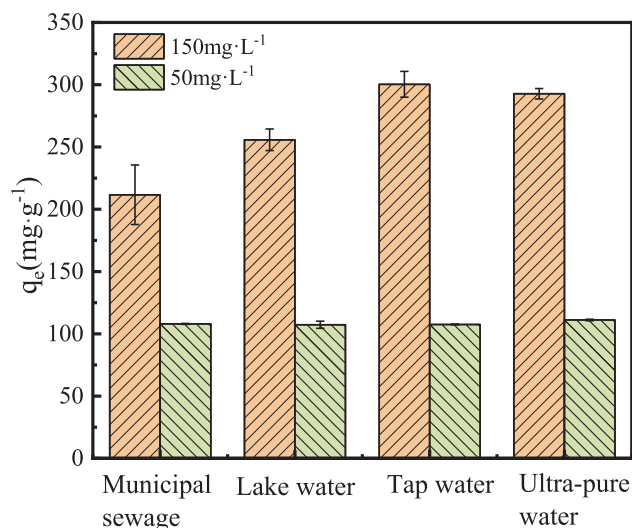


Fig. 9. TC removal in different water by adsorption on SAC-3. (Experimental conditions: $C_0 = 50 \text{ mg}\cdot\text{L}^{-1}$ and $150 \text{ mg}\cdot\text{L}^{-1}$, $m = 50 \text{ mg}$, $V = 200 \text{ mL}$, $T = 298 \text{ K}$, $\text{pH} = 6.0$ and $t = 12 \text{ h}$).

was 3:1, and the maximum surface area was $1365.79 \text{ m}^2\cdot\text{g}^{-1}$. The SAC showed high adsorption capacity ($q_m = 302.99 \text{ mg}\cdot\text{g}^{-1}$) for TC in solution. Langmuir model can fit the TC adsorption on SAC-3, and the mechanism of adsorption of TC by SAC-3 may be hydrogen bonding and π - π bond interaction. SAC-3 exhibits high affinity for aqueous TC, and Na^+ the impact on TC adsorption is small, while Ca^{2+} inhibits TC adsorption on the adsorbent. In addition, TC saturated SAC can be regenerated in alkaline solution and then reused.

Author contribution

J.H. Wang conceived the research; J.H. Wang and S.L. Lei wrote the main manuscript and prepared all the figures and tables; S.L. Lei and L.Q. Liang performed the experiments; J.H. Wang supervised the project. All authors discussed the results and agreed on the presentation of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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